

Practice Questions for 2024-11-05-NetFlows

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1 Questions

1. In the network flow problem depicted in Slide 4, what is the total supply in the network?
 - a. 0
 - b. 6
 - c. 12
 - d. 18
 - e. 24

2. In the context of minimum-cost network flow problems, what is the significance of the incidence matrix A , and how does its structure relate to the flow conservation constraints at each node?
 - a. The incidence matrix A is irrelevant to the flow conservation constraints; these constraints are determined solely by the supply/demand vector b .
 - b. The incidence matrix A represents the network's structure, with each column representing a node and each row representing an arc. The flow conservation constraints are derived from the dot product of A and the flow vector x .
 - c. The incidence matrix A represents the network's structure, with each row representing a node and each column representing an arc. The flow conservation constraints are directly encoded in the rows of A , with each row summing to zero.
 - d. The incidence matrix A represents the network's structure, with each row representing a node and each column representing an arc. The flow conservation constraints are derived from the equation $Ax = b$, where x is the flow vector and b is the supply/demand vector.
 - e. The incidence matrix A is a square matrix representing the connectivity of the network. The flow conservation constraints are determined by the eigenvalues of A .

3. Why is the rank of the incidence matrix A in a network flow problem equal to $m - 1$, where m is the number of nodes, and what are the implications of this for solving the problem using linear programming?
 - a. The rank of A is $m - 1$ because there are $m - 1$ independent flow conservation equations; one equation is redundant due to the overall balance of supply and demand.
 - b. The rank of A is $m - 1$ because the incidence matrix is always singular; this means the network flow problem is always infeasible.
 - c. The rank of A is $m - 1$ because the network must be connected; otherwise, the problem is not well-defined.

- d. The rank of A is $m - 1$ due to the inherent symmetry of the network; this symmetry simplifies the solution process.
 - e. The rank of A being $m - 1$ is a coincidence and has no particular significance for solving the problem.
4. Explain the relationship between spanning trees in a network and bases in the simplex method when solving minimum-cost network flow problems. How does this relationship facilitate the solution process?
- a. There is no relationship between spanning trees and simplex bases; they are entirely separate concepts.
 - b. A spanning tree corresponds to a basis in the simplex method, where the arcs in the tree represent basic variables and the arcs not in the tree represent non-basic variables. This allows for efficient iterative solution using the simplex method.
 - c. Spanning trees are used to visualize the network, but they have no direct connection to the simplex method's solution process.
 - d. A spanning tree represents a feasible solution, but not necessarily a basic solution, in the simplex method. The simplex method is used to find a spanning tree that corresponds to an optimal solution.
 - e. Spanning trees are only relevant for solving maximum flow problems, not minimum-cost flow problems.
5. In Slide 2, what is the total supply in the network?
- a. 6
 - b. 12
 - c. 18
 - d. 24
 - e. 0
6. In Slide 4, what is the total demand in the network?
- a. 6
 - b. 12
 - c. 18
 - d. 24
 - e. 0
7. Referring to Slide 5, what is the entry $A_{a,(a,e)}$ in the incidence matrix A ?
- a. 1
 - b. -1
 - c. 0
 - d. 2

- e. It is undefined
8. In Slide 7, what is the dual problem's objective function?
- a. $c^T x$
 - b. $-b^T y$
 - c. $b^T y$
 - d. $Ax = -b$
 - e. $A^T y \leq c$
9. In the context of minimum-cost network flow problems, what is the significance of the incidence matrix A , and how does its structure relate to the flow conservation constraints at each node?
- a. The incidence matrix A is irrelevant to the flow conservation constraints; they are handled separately.
 - b. The incidence matrix A represents the network's structure, and each row's sum, when multiplied by the flow vector, represents the net flow into or out of the corresponding node.
 - c. The incidence matrix A is a square matrix whose determinant determines the feasibility of the network flow problem.
 - d. The incidence matrix A is used only in the dual formulation of the network flow problem, not the primal.
 - e. The incidence matrix A is a diagonal matrix whose diagonal entries represent the capacity of each arc in the network.
10. In Slide 4, what is the total demand in the network?
- a. 6
 - b. 11
 - c. 17
 - d. 23
 - e. 0

2 Answers

1. In the network flow problem depicted in Slide 4, what is the total supply in the network?
 - a. The total supply is calculated by summing the positive values of b_i (supply) and subtracting the absolute values of negative b_i (demand). In Slide 4, the supply values are 8 and 4, and the demand values are -2, -6, -6, and 5. Therefore, the total supply is $8 + 4 - |-2| - |-6| - |-6| + |5| = 12 - 2 - 6 - 6 + 5 = 3$. However, the total supply must equal the total demand for a balanced network flow problem. Summing all b_i values gives $8 + 0 + (-2) + (-6) + 0 + 4 + (-5) = -1$. There must be an error in the slide's depiction of the supply/demand values. If we assume the total supply equals the total demand, then the total supply is 0.
 - b. Incorrect. This is not the correct sum of supplies and demands.
 - c. Incorrect. This is not the correct sum of supplies and demands.
 - d. Incorrect. This is not the correct sum of supplies and demands.
 - e. Incorrect. This is not the correct sum of supplies and demands.

2. In the context of minimum-cost network flow problems, what is the significance of the incidence matrix A , and how does its structure relate to the flow conservation constraints at each node?
 - a. The supply/demand vector b provides the net supply or demand at each node, but the incidence matrix A is crucial for defining how flows on arcs affect the node balances.
 - b. The statement is incorrect because the columns of A represent arcs, not nodes, and the rows represent nodes.
 - c. While each row of A represents a node, the rows do not necessarily sum to zero. The flow conservation constraint is expressed as a balance equation for each node, not simply a row sum.
 - d. The incidence matrix A indeed represents the network's structure, with each row corresponding to a node and each column to an arc. The equation $Ax = b$ directly encodes the flow conservation constraints, where each row represents the balance of inflow and outflow at a particular node.
 - e. The incidence matrix is not necessarily square, and its eigenvalues are not directly related to the flow conservation constraints.

3. Why is the rank of the incidence matrix A in a network flow problem equal to $m - 1$, where m is the number of nodes, and what are the implications of this for solving the problem using linear programming?
 - a. The rank deficiency arises because the sum of all rows in the incidence matrix is always a zero vector, reflecting the overall balance of supply and demand in the network. This redundancy means one flow conservation equation can be removed without losing information, simplifying the linear program.
 - b. The singularity of A does not imply infeasibility. It simply means that one of the flow conservation equations is linearly dependent on the others.
 - c. Connectivity is important for the problem to be meaningful, but it doesn't directly explain why the rank is $m - 1$.
 - d. While network structures can exhibit symmetry, this is not the fundamental reason for the rank deficiency of the incidence matrix.
 - e. The rank of the incidence matrix is a crucial property that significantly impacts the solution process of network flow problems.

4. Explain the relationship between spanning trees in a network and bases in the simplex method when solving minimum-cost network flow problems. How does this relationship facilitate the solution process?
 - a. The connection between spanning trees and simplex bases is a cornerstone of efficient network flow algorithms.
 - b. This option correctly describes the fundamental relationship. A spanning tree defines a basic feasible solution, where the flows on arcs outside the tree are zero (non-basic variables). The simplex method iteratively improves this solution by changing the spanning tree (basis).
 - c. Spanning trees are not merely visual aids; they are directly linked to the algebraic structure of the problem and the simplex method's iterative process.
 - d. A spanning tree does correspond to a basic solution, and the simplex method works by changing the spanning tree to improve the solution.
 - e. Spanning trees are fundamental to solving both maximum flow and minimum-cost flow problems efficiently.

5. In Slide 2, what is the total supply in the network?
 - a. This is incorrect. It only considers the supply at node f.
 - b. The total supply is the sum of all positive b_i values. In Slide 2, we have $b_a = 8$, $b_f = 4$. Therefore, the total supply is $8 + 4 = 12$.
 - c. This is incorrect. It is not the sum of all supplies.
 - d. This is incorrect. It is not the sum of all supplies.
 - e. While the total supply equals the total demand, the question asks for the total supply alone. The total supply is not 0.

6. In Slide 4, what is the total demand in the network?
 - a. This is incorrect. It only considers the demand at node c.
 - b. This is incorrect. It is not the sum of all demands.
 - c. The total demand is the sum of the absolute values of all negative b_i values. In Slide 4, we have $b_c = -2$, $b_d = -6$, $b_g = -5$. Therefore, the total demand is $|-2| + |-6| + |-5| = 13$.
 - d. This is incorrect. It is not the sum of all demands.
 - e. While the total supply equals the total demand, the question asks for the total demand alone. The total demand is not 0.

7. Referring to Slide 5, what is the entry $A_{a,(a,e)}$ in the incidence matrix A?
 - a. This would be true if node a were the head of arc (a,e).
 - b. According to the definition in Slide 5, $A_{i,(k,l)} = -1$ if $i = k$, $A_{i,(k,l)} = 1$ if $i = l$, and $A_{i,(k,l)} = 0$ otherwise. Since node a is the tail of arc (a,e), the entry is -1.
 - c. This would be true if node a were neither the head nor tail of arc (a,e).
 - d. There is no such entry in the incidence matrix.
 - e. The entry is well-defined according to the definition given in Slide 5.

8. In Slide 7, what is the dual problem's objective function?
- This is the primal objective function.
 - The dual problem in Slide 7 is given as $\max -b^T y$ subject to $A^T y \leq c$. Therefore, the objective function is $-b^T y$.
 - This is the negative of the dual objective function.
 - This is a constraint in the primal problem.
 - This is a constraint in the dual problem.
9. In the context of minimum-cost network flow problems, what is the significance of the incidence matrix A , and how does its structure relate to the flow conservation constraints at each node?
- Incorrect. The incidence matrix is fundamental to representing the flow conservation constraints. The structure of A directly reflects the connections between nodes and how flow is balanced.
 - The incidence matrix A encodes the network's topology. Each row corresponds to a node, and the entries in that row indicate the arcs entering and leaving that node. When multiplied by the flow vector, the row sum represents the net flow into or out of the node, which must equal the node's supply or demand for flow conservation.
 - Incorrect. The incidence matrix is not square; it has dimensions (number of nodes) $\times$ (number of arcs). Its determinant is not defined, and feasibility is determined by the existence of a non-negative solution to the system $Ax = -b$.
 - Incorrect. The incidence matrix A is central to both the primal and dual formulations of the network flow problem. It appears in both $Ax = -b$ (primal) and $A^T y \leq c$ (dual).
 - Incorrect. The incidence matrix A is not a diagonal matrix. Diagonal entries would not represent arc capacities; rather, they would represent the net supply/demand at each node.
10. In Slide 4, what is the total demand in the network?
- The sum of negative values in the b_i vector is not 6.
 - The total demand is the sum of the negative values in the b_i vector shown in Slide 4. Adding the negative values $(-2) + (-6) + (-6) + (-2) + (-5) = -21$. However, the total supply is 23 ($8 + 4 + 9 + 2$). Since the network is balanced, the total demand must be equal to the total supply, which is 23. Therefore, the total demand is 23. There seems to be an error in the slide, as the sum of the values shown does not equal zero. The correct answer should be 23, reflecting the balanced nature of the network.
 - The sum of negative values in the b_i vector is not 17.
 - The sum of negative values in the b_i vector is not 23. The question asks for the total demand, which is the sum of the negative values. The sum of the negative values is -11. However, since the network is balanced, the total demand should equal the total supply. There is an error in the slide's values.
 - The network is not balanced, meaning the total supply does not equal the total demand. The total demand is not 0.